which can be equivalently written as

$$\begin{bmatrix}\begin{bmatrix}0\\ V_{\phi,\mathrm{tr}}\otimes I_{n_{\mathrm{x}}n_\phi}\end{bmatrix}^{\mathsf{H}}\\ I\end{bmatrix}^{\mathsf{H}} \times \begin{bmatrix}-I & \begin{bmatrix}0\\ \hat{V}_\phi\otimes I_{n_{\mathrm{x}}n_\phi}\end{bmatrix}^{\mathsf{H}}\\ \begin{bmatrix}0\\ \hat{V}_\phi\otimes I_{n_{\mathrm{x}}n_\phi}\end{bmatrix} & \begin{bmatrix}0 & 0\\ 0 & (\tilde{\Gamma}_\phi-\hat{V}_\phi\hat{V}_\phi^{\mathsf{H}})\otimes I_{n_{\mathrm{x}}n_\phi}\end{bmatrix}\end{bmatrix} \times \begin{bmatrix}\begin{bmatrix}0\\ V_{\phi,\mathrm{tr}}\otimes I_{n_{\mathrm{x}}n_\phi}\end{bmatrix}^{\mathsf{H}}\\ I\end{bmatrix} \succeq 0. \tag{C.6}$$

By using the matrix S-lemma, Inequality (C.4) holds for all $V_{\phi,\mathrm{tr}}$ satisfying Inequality (C.6), if $S_{\text{exp-2}}(\epsilon,\tau_2,U_{\mathrm{e}},\hat{Z},\bar{W}_Z,\hat{V}_\phi,\tilde{\Gamma}_\phi,\tilde{D}_{\mathrm{des}},\bar{D}_2)\succeq 0$ (49b) holds with $\tau_2\geq 0$.

Part III. Inequality (48c) can be written as

$$\begin{bmatrix}\begin{bmatrix}V_{\phi,\mathrm{tr}}\\ 0\end{bmatrix}^{\mathsf{H}}\\ I\end{bmatrix}^{\mathsf{H}} \times \begin{bmatrix}(1-\epsilon)U_{\mathrm{e}}U_{\mathrm{e}}^T & 0\\ 0 & \begin{bmatrix}-\left(\frac{1-\epsilon}{\epsilon}\right)\bar{W}_\phi-\frac{\gamma_{\mathrm{w}}}{T}D_{\mathrm{des}} & 0\\ 0 & 0\end{bmatrix}-\bar{D}_3\end{bmatrix} \times \begin{bmatrix}\begin{bmatrix}V_{\phi,\mathrm{tr}}\\ 0\end{bmatrix}^{\mathsf{H}}\\ I\end{bmatrix} \succeq 0. \tag{C.7}$$

From (36) and (38), we have

$$V_{\phi,\mathrm{tr}}V_{\phi,\mathrm{tr}}^{\mathsf{H}}-V_{\phi,\mathrm{tr}}\hat{V}_\phi^{\mathsf{H}}-\hat{V}_\phi V_{\phi,\mathrm{tr}}^{\mathsf{H}}+\hat{V}_\phi\hat{V}_\phi^{\mathsf{H}}\preceq\tilde{\Gamma}_\phi, \tag{C.8}$$

which can be equivalently written as

$$\begin{bmatrix}\begin{bmatrix}V_{\phi,\mathrm{tr}}\\ 0\end{bmatrix}^{\mathsf{H}}\\ I\end{bmatrix}^{\mathsf{H}} \begin{bmatrix}-I & \begin{bmatrix}\hat{V}_\phi\\ 0\end{bmatrix}^{\mathsf{H}}\\ \begin{bmatrix}\hat{V}_\phi\\ 0\end{bmatrix} & \begin{bmatrix}(\tilde{\Gamma}_\phi-\hat{V}_\phi\hat{V}_\phi^{\mathsf{H}}) & 0\\ 0 & 0\end{bmatrix}\end{bmatrix} \times \begin{bmatrix}\begin{bmatrix}V_{\phi,\mathrm{tr}}\\ 0\end{bmatrix}^{\mathsf{H}}\\ I\end{bmatrix} \succeq 0. \tag{C.9}$$

By using the matrix S-lemma, Inequality (C.7) holds for all $V_{\phi,\mathrm{tr}}$ satisfying Inequality (C.9), if the following holds with $\tau_3\geq 0$:

$$\begin{bmatrix}(1-\epsilon)U_{\mathrm{e}}U_{\mathrm{e}}^T & 0\\ 0 & \begin{bmatrix}-\left(\frac{1-\epsilon}{\epsilon}\right)\bar{W}_\phi-\frac{\gamma_{\mathrm{w}}}{T}D_{\mathrm{des}} & 0\\ 0 & 0\end{bmatrix}-\bar{D}_3\end{bmatrix} -\tau_3\begin{bmatrix}-I & \begin{bmatrix}\hat{V}_\phi\\ 0\end{bmatrix}^{\mathsf{H}}\\ \begin{bmatrix}\hat{V}_\phi\\ 0\end{bmatrix} & \begin{bmatrix}(\tilde{\Gamma}_\phi-\hat{V}_\phi\hat{V}_\phi^{\mathsf{H}}) & 0\\ 0 & 0\end{bmatrix}\end{bmatrix} \succeq 0. \tag{C.10}$$

From Lemma 12, we have

$$U_{\mathrm{e}}U_{\mathrm{e}}^T\succeq U_{\mathrm{e}}\hat{U}^\top+\hat{U}U_{\mathrm{e}}^\top-\hat{U}\hat{U}^\top. \tag{C.11}$$

Inserting Inequality (C.11) in Inequality (C.10) yields $S_{\text{exp-3}}(\epsilon,\tau_3,U_{\mathrm{e}},\hat{U},\hat{W}_\phi,\hat{V}_\phi,\tilde{\Gamma}_\phi,D_{\mathrm{des}},\bar{D}_3)\succeq 0$ (49c).

Therefore, if there exist matrices U_{e}, $\bar{D}_1$, $\bar{D}_2$ and $\bar{D}_3$ that satisfy Inequalities (49a), (49b), (49c) with $\bar{D}_3$, with $\bar{D}_1+\bar{D}_2+\bar{D}_3\succeq 0$ (cf. (50)), then Inequality (45) is satisfied for all $V_{\mathrm{x,tr}}$, $V_{\phi,\mathrm{tr}}$, $Y_{\mathrm{x,tr}}$, and $Y_{\phi,\mathrm{tr}}$ satisfying bounds (38), (39), and the exploration goal (4) is achieved. □

D Sample-based constants $\tilde{\Gamma}_{\mathrm{x}}$, Γ_{x} and Γ_ϕ

Sample-based bounds $\tilde{\Gamma}_{\mathrm{x}}$, $\tilde{\Gamma}_\phi$ (38) and Γ_{x}, Γ_ϕ (39) can be determined using the 'scenario' approach [6] .

D.1 Scenario approach to estimate $\tilde{\Gamma}_{\mathrm{x}}$ and $\tilde{\Gamma}_\phi$

The bounds $\tilde{\Gamma}_{\mathrm{x}}$ and $\tilde{\Gamma}_\phi$ are computed under the assumption that $\theta_{\mathrm{tr}}\in\mathbf{\Theta}_0$ (cf. Assumption 2). The bounds $\tilde{\Gamma}_{\mathrm{x}}$